

Pre-Algebra Team Exam

2026 James Clemens Math Tournament

INSTRUCTIONS:

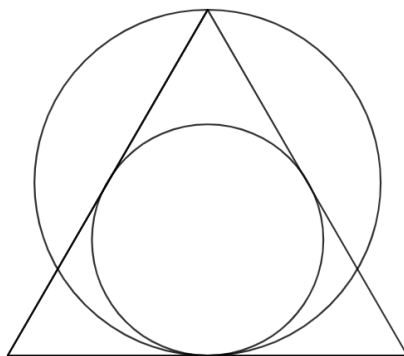
1. Your team has 60 minutes to complete this examination.
2. One official team from each school consisting of four students and other unofficial teams may compete. Only the scores from the official team will be counted towards a school's final score.
3. This exam consists of 15 free-response questions, each worth 10 points if answered correctly and 0 points if answered incorrectly or left unanswered.
4. Calculators, cellular devices, books, notes, and other aids are prohibited during this exam. Scratch paper will be provided for calculations. If you require additional scratch paper, simply raise your hand and your proctor will assist you.
5. Write your answers to the questions in the provided team answer sheet. You may use the test booklets for scratch work, but only the answers marked in the team answer sheet will be counted. Please write **legibly**.
6. **All answers to the team exam will be a positive integer from 1 to 999, inclusive.**
7. **Leave out** all units and special characters when recording your answer on the team answer sheet. For example, if you get \$1 as your answer, write 1 on the answer sheet.
8. A school's final score for a grade will be calculated by adding the top 4 written test scores in that grade (excluding tiebreaker questions) as well as the team round score of the official team, for a maximum of 550 points.
9. In the event of a tie, the school with the highest team round score will be favored. If team round scores result in another tie, answers will be evaluated starting backwards from question 15 to 1 to determine a winner.
10. Diagrams are not necessarily drawn to scale.
11. Although this math tournament is intended to demonstrate your knowledge and skills in math, be sure to enjoy the experience and have fun!
12. You may keep the test booklet after the exam.

DO NOT TURN THIS PAGE UNTIL TOLD TO DO SO

1. Let $A\clubsuit B$ be equal to $A^3 - \sqrt{B}$ and $C\star D$ be equal to $2C + D$. Find $\frac{(3\clubsuit 4)\star 14}{(2\clubsuit 9)\star 6}$.
2. Carl's house and Nick's house are located at the ends of a 462 mile road. Every 6 miles, a toll fee of \$2.50 must be paid. If there are no tolls at Carl's house or Nick's house, how much does it cost, in dollars, for Carl to drive to Nick's house?
3. Charles and Michael are mowing lawns. Charles can mow 1 lawn in 30 minutes. Michael can mow 5 lawns in 2 hours. Working together, how many minutes will it take to mow 18 lawns?
4. How many ways are there to rearrange the letters in the word *PEOPLE*?
5. Solve for x : $\sqrt{\frac{x}{\sqrt{\frac{x}{\sqrt{x}}}}} = 8$.
6. A line contains the points $(2, 7)$ and $(-1, -2)$. If the equation for the line can be expressed as $y = mx + b$, find $\frac{m+b}{m-b}$.
7. Joel watched 15 episodes of his favorite TV show in one sitting. If each episode is 28 minutes long, how many hours did he watch his favorite TV show?
8. Two cars start at the same location and are driving away from each other. Car A travels north at 3 miles per hour. Car B travels east at 4 miles per hour. After 2 hours, how far apart are the two cars, in miles?

9. Find the value of the hundredths place of $(M - A)/(T + H)$ if:
- M is the sum of the exterior angles of any polygon.
 - A is the numerator of the probability, expressed as a fraction in simplest form, that the sum of rolling two standard six-sided dice is greater than or equal to 6.
 - T is the sum of the first 10 positive integers.
 - H is the sum of the interior angles of a regular heptagon.

10. The following figure shows a circle inscribed in an equilateral triangle. There is also a larger circle that is equal in height to the equilateral triangle, as shown in the figure below. If the area of the smaller circle is 16π , and the area of the larger circle can be expressed as $x\pi$, find x .



11. The letters *PREALGEBRA* are placed in a box. If the probability that you draw the letters A , E , and G out of the box without replacement in that order is expressed as $\frac{x}{y}$ where x and y are relatively prime, find $x + y$.
12. In music, the number of prefixes meaning “half” in the name of a note correlates to the number of them in one beat. For example, there are 2^4 semiquavers in one beat and 2^5 demisemiquavers in one beat. How many hemidemisemihemidemisemiquavers are in one beat? (Note: The prefixes “hemi”, “demi”, and “semi” all mean “half” in this context.)
13. An equilateral triangle has a side length 5. If the area of the triangle can be expressed as $\frac{x\sqrt{y}}{z}$ where x and z are relatively prime, find $\frac{x - y}{0.5z}$.

14. If (x, y, z) is the solution to the system of equations below, find $\frac{xz}{y}$:

$$\begin{cases} x + y + z = 35 \\ x + 2y + 3z = 77 \\ 3x - 2y - z = -15 \end{cases}$$

15. The 1^{st} term in an arithmetic sequence is 2 and the 3^{rd} term is 8. Find the sum of all the terms up until the 10^{th} term (including the 10^{th} term).